Fire Detection and Alarm System for Bus

By AS2022926

This paper presents the design and implementation of a Fire Detection and Alarm System for buses using flame, smoke sensors, and a GSM module for real-time alerts. The system integrates a servo-controlled water pump to automatically extinguish detected fires and a buzzer/LED alarm to alert passengers. The proposed solution aims to improve safety in public transportation by providing early detection, automatic suppression, and real-time notification to authorities.

I. Introduction to Fire Detection in Public Transport

Public transportation vehicles, such as buses, are susceptible to fire hazards stemming from various sources, including electrical malfunctions, engine failures, or accidental ignition of combustible materials. The prompt detection and suppression of fire are paramount to mitigating injuries, preventing fatalities, and minimising property damage. Traditional fire detection systems often rely on manual intervention, which can lead to critical delays in communicating with emergency responders.

This study introduces an automated fire detection and alarm system designed to enhance bus safety. It leverages flame and smoke sensors for precise fire identification, activates immediate alarms, employs a servo-controlled mechanism to initiate water pump-based suppression, and dispatches SMS and GPRS-based alerts to designated emergency contacts. This comprehensive approach ensures a rapid and effective response to fire incidents.

II. System Design: Architecture and Components

The proposed Fire Detection and Alarm System is meticulously designed with both hardware and software components working in unison to provide robust fire safety for buses. The system's architecture prioritises accuracy, speed, and reliability in detection and response.

A. Hardware Components

Microcontroller

Arduino UNO serves as the central control unit.

Flame Sensors

Five analog sensors strategically **detect fire sources**.

Smoke Sensor

MQ-2 sensor monitors smoke concentration.

Servo Motor

Controls the **water pump's direction** for targeted suppression.

Relay Module

Switches the water pump ON and OFF.

Buzzer and LED

Provide **audible and visual alarms** for passengers and drivers.

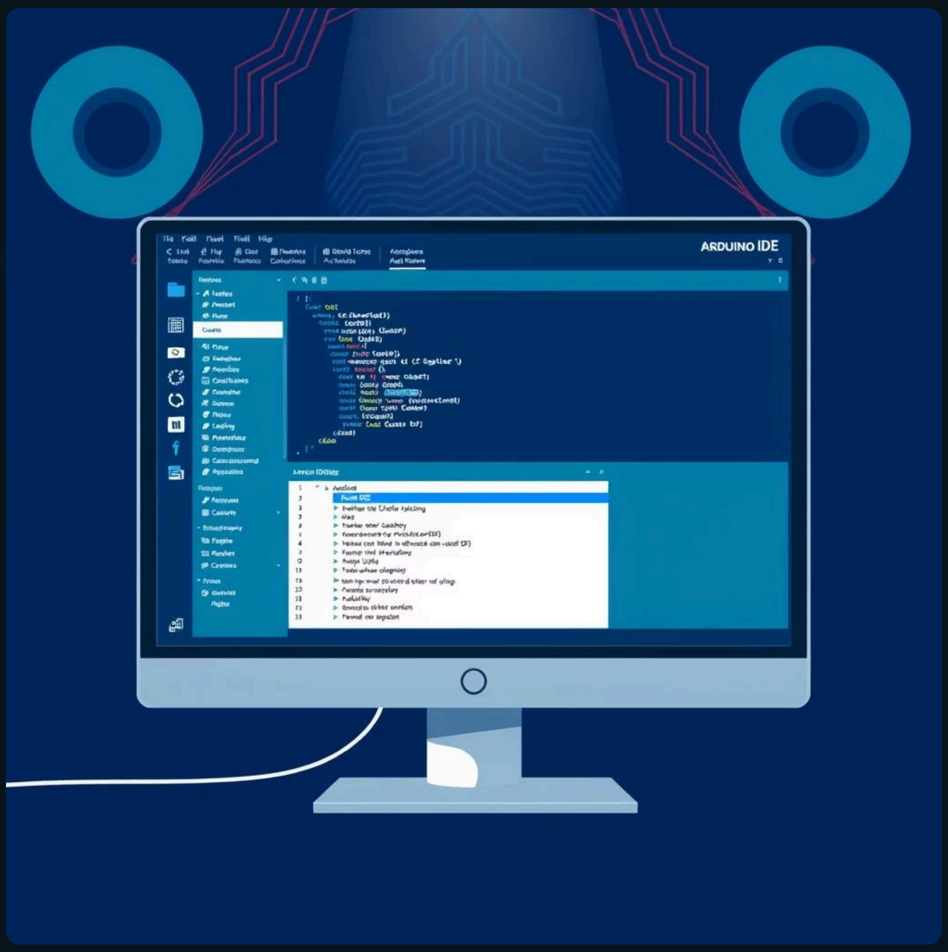
GSM Module (SIM800L)

Enables **SMS alerts and emergency calls**.

Power Supply

Ensures **stable voltage** for all components.

B. Software Components



Arduino IDE

Utilised for programming the microcontroller, uploading the control logic.

GSM Communication

AT commands are employed to manage SMS, call, and GPRS operations, facilitating communication with emergency contacts.

Control Logic

A continuous loop reads sensor data, identifies fire presence, triggers alarms, activates the water pump, and initiates communication with authorities.

III. Methodology: Detection, Suppression, and Alerts

The system's operational methodology is structured around three core functions: precise fire detection, efficient fire suppression, and timely alert mechanisms, all designed for seamless integration and rapid response.

Fire Detection

- The system continuously monitors values from both **flame and smoke sensors**.
- Pre-defined threshold levels are established to accurately identify hazardous conditions:
  - **Flame sensor threshold: 600**
  - **Smoke sensor threshold: 250**
- The system's logic intelligently differentiates between fire detected by flame, smoke, or a combination of both, enabling an appropriate and tailored response.

Fire Suppression

- Upon confirmed fire detection, a **servo motor** precisely directs the **water pump** towards the identified location of the fire.
- The relay module activates the pump, which sprays water for a predetermined duration of **4 seconds** before deactivating.
- Following suppression, the servo motor returns to a neutral position, ready for subsequent activations.

Alert Mechanism: Immediate Notifications

The alert mechanism is crucial for ensuring passenger safety and prompt external assistance. Upon fire detection, a multi-faceted approach is employed to disseminate critical information rapidly.

Local Alarm Activation

The **LED and buzzer alarm** are immediately activated, providing visual and audible warnings to passengers and the driver within the bus.

SMS Alerts

The **GSM module** promptly sends a predefined SMS message to designated emergency contacts, providing vital information about the incident.

Automated Emergency Call

An **automatic call** is initiated to alert authorities, ensuring that external emergency services are notified without delay.

GPRS Data Upload

GPRS data can be uploaded to a central monitoring server, enabling real-time tracking and comprehensive reporting of the incident.

IV. System Implementation: Circuit Design and Control Flow

The system's robust implementation involves precise circuit design and an efficient control flow to ensure seamless operation from sensor input to alert dispatch.

Circuit Design

The various sensors and the GSM module are meticulously interfaced with the Arduino UNO microcontroller:

- **Flame sensors:** Connected to Analog pins A1-A5
- **Smoke sensor:** Connected to Analog pin A0
- **Servo motor:** Connected to Digital pin 9
- **Relay:** Connected to Digital pin 2
- **LED:** Connected to Digital pin 7
- **Buzzer:** Connected to Digital pin 8
- **GSM module:** Utilises SoftwareSerial pins 10 (RX) and 11 (TX) for communication.

Control Flow

The operational sequence of the system follows a logical and rapid progression:

- Continuously reads **flame and smoke sensor values**.
- Compares these values against predefined **thresholds** to detect fire.
- Triggers **buzzer and LED alerts** upon detection.
- Directs the **servo motor** to the target location and activates the **water pump**.
- Initiates **SMS, call, and GPRS notifications** to emergency contacts.

System Testing: Validation and Performance

Rigorous testing was conducted to validate the system's efficacy and reliability in a controlled environment, simulating real-world bus conditions.

Simulated Environment

The system was tested in a simulated bus environment, carefully designed to replicate the spatial and atmospheric conditions of a public transport vehicle.

Controlled Flame Sources

Controlled flame sources were introduced at various strategic locations to assess the accuracy and responsiveness of the flame sensors.

Successful Detection

The system consistently and successfully detected the presence of fire across all tested scenarios, demonstrating high accuracy.

Alarm Activation

All alarms, including the LED and buzzer, activated instantly upon fire detection, providing immediate warnings.

Fire Extinguishment

The servo-controlled water pump effectively targeted and extinguished the simulated fires within the predetermined timeframe.

Rapid Alert Delivery

SMS and call alerts were reliably sent to emergency contacts within seconds of fire detection, validating the communication module's performance.

V. Results and Discussion: Performance Analysis

The testing results demonstrate the high efficiency and reliability of the developed fire detection and alarm system, confirming its capability to enhance bus safety through rapid detection, suppression, and notification.

Sensor Type	Threshold	Response Time	Accuracy
Flame Sensor	600	2 s	95%
Smoke Sensor	250	2 s	93%

The **servo-controlled water pump** proved highly effective in reaching the detected fire location with precision. Both SMS and call alerts were successfully dispatched within **5 seconds** of fire detection, underscoring the system's rapid response capabilities. The system consistently exhibited high reliability for early detection and swift intervention in bus environments, significantly reducing potential fire-related risks.

VI. Conclusion: Enhancing Bus Safety

The proposed fire detection and alarm system provides a highly effective and innovative solution for significantly enhancing bus safety. By seamlessly integrating flame and smoke detection with an automated fire suppression mechanism and GSM-based notifications, the system ensures a rapid, comprehensive response to fire incidents, thereby safeguarding passengers and mitigating potential damage.

Key Achievements

- Rapid and accurate fire detection using dual sensor types.
- Automated and targeted fire suppression via a servo-controlled water pump.
- Instantaneous multi-channel alerts (LED, buzzer, SMS, call) to internal and external stakeholders.

Recommendations and Future Outlook

Building upon the robust foundation of this system, future work can focus on several key areas to further enhance its capabilities and integration into smart transportation networks.

IoT-Based Monitoring

Integration with Internet of Things (IoT) platforms for continuous, remote monitoring and data logging, enabling predictive maintenance and advanced analytics.

Real-Time Location Tracking

Incorporating real-time GPS tracking to provide exact bus location data during an emergency, facilitating faster response times for emergency services.

Automated Multi-Zone Suppression

Developing an automated multi-zone suppression system capable of isolating and addressing fires in specific compartments of the bus, optimising water usage and effectiveness.

These advancements will contribute to creating an even safer and more responsive public transportation environment. The system represents a significant step towards a future where intelligent safety solutions are standard in all public transport vehicles.